Section 4.1 - Derivatives of Powers, Exponents, and Sums

Constant Function Rule: If $f(x) = c$ where c is a constant then

$$\boxed{f'(x) = 0}$$

Power Rule: If $f(x) = x^n$, where n is a real number, then

$$\boxed{f'(x) = n x^{n-1}}$$

Does this make sense for
$f(x) = x^1$
$f'(x) = 1x^{1-1} = 1x^0 = 1$ ✓

Constant Multiple Property: If $f(x) = k \cdot g(x)$, where k is any real number, then

$$\boxed{f'(x) = k \cdot g'(x)}$$

Sum and Difference Property: If $h(x) = f(x) \pm g(x)$ where f and g are both differentiable functions, then

$$\boxed{h'(x) = f'(x) \pm g'(x)}$$

Example 1: Differentiate the following functions:

a) $f(x) = 3$ $\boxed{f'(x) = 0}$

b) $f(x) = e$ $\boxed{f'(x) = 0}$

c) $f(x) = \sqrt[5]{8}$ $\boxed{f'(x) = 0}$

d) $f(x) = x^4$

$f'(x) = 4x^{4-1}$

$\boxed{f'(x) = 4x^3}$

e) $f(x) = x^{1/2}$

$f'(x) = \frac{1}{2} x^{\frac{1}{2}-1}$

$\boxed{f'(x) = \frac{1}{2} x^{-1/2}}$

f) $f(x) = \sqrt[3]{x^5} = (x^5)^{1/3} = x^{5/3}$

$f'(x) = \frac{5}{3} x^{\frac{5}{3}-1}$

$\boxed{f'(x) = \frac{5}{3} x^{2/3}}$

g) $f(x) = \dfrac{5}{x^2}$

$= 5x^{-2}$

$f'(x) = 5(-2x^{-3})$

$\boxed{f'(x) = -10x^{-3}}$

h) $y = 3x^2 + 7x^1 - 9$

$y' = 3(2x^1) + 7(1x^0) - 0$

$\boxed{y' = 6x + 7}$

i) $m(t) = \sqrt{5}t - t^{1/2} + e$

$= \sqrt{5}\cdot t^1 - t^{1/2} + e$

$m'(t) = \sqrt{5}\cdot 1\cdot t^{1-1} - \frac{1}{2}t^{-1/2} + 0$

$= \sqrt{5} - \frac{1}{2}t^{-1/2}$

j) $k(x) = \dfrac{x^3+4x^2-17+x}{x} = \frac{x^3}{x} + \frac{4x^2}{x} - \frac{17}{x} + \frac{x}{x} = x^2 + 4x^1 - 17x^{-1} + 1$

$k'(x) = 2x + 4\cdot 1x^0 - 17\cdot -1x^{-2} + 0$

$k'(x) = 2x + 4 + 17x^{-2}$

k) $y = \dfrac{3x^2+x^4}{5\sqrt{x}} = \frac{3x^2}{5x^{1/2}} + \frac{x^4}{5x^{1/2}}$

$= \frac{3}{5}x^{2-1/2} + \frac{1}{5}x^{4-1/2}$

$= \frac{3}{5}x^{3/2} + \frac{1}{5}x^{7/2}$

$y' = \frac{3}{5}\cdot\frac{3}{2}x^{1/2} + \frac{1}{5}\cdot\frac{7}{2}x^{5/2}$

$y' = \frac{9}{10}x^{1/2} + \frac{7}{10}x^{5/2}$

Example 2: If a book is dropped from a building 400 feet tall, its height above the ground (in feet) after t seconds is given by $s(t) = 400 - 16t^2$

a) Compute $s'(t)$ and interpret.

$s'(t) = 0 - 16\cdot 2t$

$s'(t) = -32t$ ft/sec

velocity (in ft/s) of the object after t seconds.

b) Compute $s(2)$ and $s'(2)$ and interpret.

$s(2) = 400 - 16(2)^2 = 336$ ft

$s'(2) = -32(2) = -64$ ft/s

After 2 seconds, the book is 336 feet above the ground and its height above the ground is decreasing by 64 ft/s

Example 3: If $f(x) = 3x^4 - 2x^2$, where does the graph of the function have a horizontal tangent line?

$f'(x) = 0$

$f'(x) = 3\cdot 4x^3 - 2\cdot 2x$

$f'(x) = 12x^3 - 4x$

$f'(x) = 0$

$12x^3 - 4x = 0$

$4x(3x^2 - 1) = 0$

$4x = 0$ $\quad 3x^2 - 1 = 0$

$x = 0$ $\quad 3x^2 = 1$

$\sqrt{x^2} = \sqrt{1/3}$

$x = \pm\sqrt{1/3}$

Example 4: Suppose the total cost (in dollars) of producing x books is given by

$$C(x) = 0.5x^2 - 12x + 100$$

a) Find $C(15) - C(14)$ and interpret.

$C(15) = .5(15)^2 - 12(15) + 100 = 32.5$

$C(14) = .5(14)^2 - 12(14) + 100 = 30$

$C(15) - C(14) = 32.5 - 30 = \boxed{\$2.50}$ ← the exact cost of producing the 15th book.

b) Find $C'(14)$ and interpret.

$C'(x) = .5(2x) - 12 \cdot 1x^0 + 0$

$\boxed{C'(x) = x - 12}$

$C'(14) = 14 - 12 = \boxed{\$2/\text{book}}$

when 14 books are produced, the total cost is increasing by \$2/book

Marginal Business Functions: Approximate change in the dependent variable (cost, revenue, profit) when the independent variable (the number of items produced/sold) is changed by a single unit.

- Marginal Cost Function

  $MC(x) = C'(x)$

- Marginal Revenue Function

  $MR(x) = R'(x)$

- Marginal Profit Function

  $MP(x) = P'(x)$

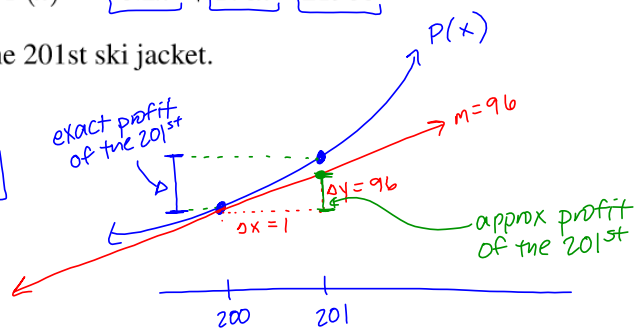
Example 5: The total profit (in dollars) of producing x ski jackets is given by

$$P(x) = -0.2x^2 + 176x - 21900$$

a) Find the exact profit realized from the sale of the 201st ski jacket.

$P(201) - P(200)$

$= 5395.8 - 5300 = \boxed{\$95.80}$



b) Use the marginal profit to approximate the profit realized from the sale of the 201st ski jacket.

Use MP(200) to approx the 201st

$MP(x) = P'(x) = -.4x + 176$

$MP(200) = -.4(200) + 176 = \boxed{\$96}$

Derivatives of Exponential and Logarithmic Functions

- If $f(x) = b^x$ then $\boxed{f'(x) = (\ln b) \cdot b^x}$
- If $f(x) = e^x$ then $\boxed{f'(x) = e^x}$
- If $f(x) = \ln x$ then $\boxed{f'(x) = \frac{1}{x}}$
- If $f(x) = \log_b x$ then $\boxed{f'(x) = \frac{1}{\ln b} \cdot \frac{1}{x}}$

x^2 vs. 2^x

power function — exp. function

Example 6: Find the derivative of each of the following functions:

a) $f(x) = 2e^x$ $\boxed{f'(x) = 2e^x}$

b) $f(x) = 7(5)^x$ $\boxed{f'(x) = 7 \cdot \ln(5) \cdot 5^x}$

c) $f(x) = \ln x + 7$

$f'(x) = \frac{1}{x} + 0$

$\boxed{f'(x) = \frac{1}{x}}$

$\log(AB) = \log(A) + \log(B)$

$\log\left(\frac{A}{B}\right) = \log(A) - \log(B)$

$\log(A^B) = B \cdot \log(A)$

d) $h(x) = 4x^2 - \sqrt[3]{x} + \log_7 x^3 - 5^x$

$= 4x^2 - x^{1/3} + 3\log_7 x - 5^x$

$\boxed{h'(x) = 8x - \frac{1}{3}x^{-2/3} + 3 \cdot \frac{1}{\ln 7} \cdot \frac{1}{x} - \ln 5 \cdot 5^x}$

e) $y = \ln(x^7) + 3(2)^x$

$= 7\ln(x) + 3(2)^x$

$\boxed{y' = 7 \cdot \frac{1}{x} + 3 \cdot \ln 2 \cdot 2^x}$

Example 7: Find the equation of the line tangent to the graph of $f(x) = e^x + \ln x$ at $x = 1$.

slope $= f'(1)$

y-coordinate of my point $= f(1)$

$f'(x) = e^x + \frac{1}{x}$

$f'(1) = e^1 + \frac{1}{1}$

$= e + 1$

$f(1) = e^1 + \ln(1)$

$= e + 0$

$= e$

$(1, e)$ $m = e + 1$

$y = mx + b$

$e = (e+1)(1) + b$

$\cancel{e} = \cancel{e} + \cancel{1} + b$

$-1 \; -\cancel{e} \; -\cancel{e} \; -\cancel{1}$

$-1 = b$

$\boxed{y = (e+1)x - 1}$

Section 4.1 Highly Suggested Homework Problems: 1-75(odd), 79, 97